MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 10** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. If TP and TQ are two tangents drawn from T to a circle with centre O such that $\angle PTQ = 70°$, then $\angle OPQ$ is equal to: [1]

(A) 35° (B) 45° (C) 55° (D) 70°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) 35°
Justification: Since $\angle PTQ = 2\angle OPQ$, $\angle OPQ = \frac{1}{2}(70°) = \boxed{35°}$.

Q2. In a right triangle ABC with $\angle B = 90°$, $AB = 8$ cm and $BC = 6$ cm. The radius of the circle inscribed in the triangle is: [1]

(A) 1.5 cm (B) 2 cm (C) 2.5 cm (D) 3 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 2 cm
Justification: Hypotenuse $AC = 10$ cm. Inradius $r = \frac{8+6-10}{2} = \boxed{2 \text{ cm}}$.

Q3. If a chord of the outer circle of two concentric circles has length 14 cm and is tangent to the inner circle of radius 24 cm, the outer radius is: [1]

(A) 25 cm (B) 26 cm (C) 28 cm (D) 31 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) 25 cm
Justification: Half-chord = 7 cm, inner radius = 24 cm. Outer radius $R = \sqrt{24^2 + 7^2} = \sqrt{625} = \boxed{25 \text{ cm}}$.

Q4. If PA and PB are tangents to a circle from P such that $PA = 9$ cm and $\angle APB = 60°$, then the length of chord AB is: [1]

(A) 4.5 cm (B) 9 cm (C) $9\sqrt{3}$ cm (D) 6 cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 9 cm

Justification: $PA = PB = 9$ cm and $\angle APB = 60° \implies \triangle PAB$ is equilateral, so $AB = \boxed{\mathbf{9}\text{ cm}}$.

Q5. The maximum number of parallel tangents that a single circle can have at most is: [1]

(A) 1 (B) 2 (C) 3 (D) Infinite

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 2

Justification: For any fixed line orientation (secant/tangent slope), a circle has at most $\boxed{\mathbf{2}}$ parallel tangents (at opposite ends of the perpendicular diameter).

Q6. **Assertion (A):** If a right triangle ABC with sides 3 cm, 4 cm, 5 cm circumscribes a circle, then the radius of the incircle is 1 cm.
Reason (R): The inradius r of a right triangle with legs a, b and hypotenuse c is given by $r = \frac{a+b-c}{2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Applying Reason (R) with $a = 3, b = 4, c = 5$ directly yields $r = \frac{3+4-5}{2} = 1$ cm. Both are true and (R) explains (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | ($2 \times 2 = 4$ Marks)

Q7. The tangent at a point C of a circle and a diameter AB when extended intersect at P. If $\angle PCA = 110°$, find $\angle CBA$. [2]

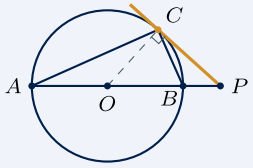
► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Join OC. By Theorem 10.1, $OC \perp CP \implies \angle OCP = 90°$.

Step 1: Finding $\angle OCA$ and $\angle OAC$ [1 Mark]
$\angle OCA = \angle PCA - \angle OCP = 110° - 90° = 20°$.
Since $OA = OC$ (radii), $\angle CAB = \angle OCA = 20°$.

Step 2: Angle Sum in Semicircle $\triangle ABC$ [1 Mark]
Since AB is a diameter, $\angle ACB = 90°$. In $\triangle ABC$:

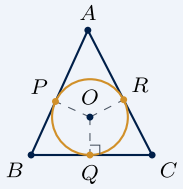
$$\angle CBA = 180° - 90° - 20° = \boxed{\mathbf{70°}}$$



Q8. A circle is inscribed in $\triangle ABC$ touching AB, BC, CA at P, Q, R. If $AB = 12$ cm, $BC = 8$ cm, and $AC = 10$ cm, find the length of AD. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

(Note: Point D denotes touchpoint P on side AB, following standard textbook labelling.)



Step 1: Equal Tangent Relations [1 Mark]
Let $AP = AR = x$, $BP = BQ = y$, $CQ = CR = z$. Then $x + y = 12$, $y + z = 8$, $z + x = 10$. Adding gives $2(x + y + z) = 30 \implies x + y + z = 15$ cm.

Step 2: Finding Tangent Segments [1 Mark]
AP (or AD) $= 15 - 8 = \mathbf{7}$ cm (and $BP = 5$ cm, $CQ = 3$ cm).

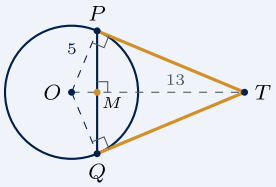
$$\boxed{AP = AD = \mathbf{7} \text{ cm}}$$

SECTION C: Short Answer Type — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. From a point T outside a circle with centre O, tangents TP and TQ are drawn. If $OT = 13$ cm and radius is 5 cm, find the length of chord PQ. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Join OT intersecting chord PQ perpendicularly at midpoint M.



Step 1: Length of Tangent TP [1 Mark]
In right $\triangle OPT$: $TP = \sqrt{OT^2 - OP^2} = \sqrt{13^2 - 5^2} = 12$ cm.

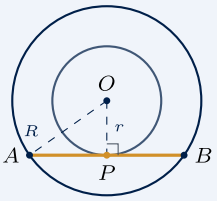
Step 2: Finding Half-Chord PM via Area [2 Marks]
Area$(\triangle OPT) = \frac{1}{2}(OP)(TP) = \frac{1}{2}(OT)(PM) \implies 5 \times 12 = 13 \times PM \implies PM = \frac{60}{13}$ cm.

$$\boxed{\text{Chord } PQ = 2PM = \frac{\mathbf{120}}{\mathbf{13}} \text{ cm} \quad (\approx \mathbf{9.23} \text{ cm})}$$

Q10. Prove that in two concentric circles, all chords of the larger circle which touch the smaller circle are of equal length. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let outer circle have radius R, inner circle radius r $(R > r)$, and AB be any chord touching the inner circle at P.



Step 1: Chord Bisector Property [1.5 Marks]
Since AB is tangent to the inner circle at P, $OP \perp AB$ (Theorem 10.1), and since $OP \perp AB$, P bisects chord AB $(AP = PB)$.

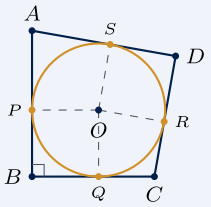
Step 2: Constant Chord Length [1.5 Marks]
In right $\triangle OPA$, $AP = \sqrt{OA^2 - OP^2} = \sqrt{R^2 - r^2}$. Thus $AB = 2\sqrt{R^2 - r^2}$, which depends solely on the fixed radii R, r.

$$\boxed{\therefore \text{ All such tangent chords are equal in length [Proved]}}$$

Q11. In the given figure, a circle touches all four sides of a quadrilateral $ABCD$ whose sides are AB, BC, CD, DA. (i) Prove that $AB + CD = AD + BC$. (ii) If $AB = 12$, $BC = 15$, $CD = 14$ cm, find DA. (iii) If $\angle B = 90°$ and $r = 4$ cm, find the lengths of the tangent segments on side AB. [5]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let the circle touch AB, BC, CD, DA at P, Q, R, S.



(i) Proof of Pitot Relation: [2 Marks]

By Theorem 10.2: $AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$.

Adding: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \implies AB + CD = AD + BC$. [Proved]

(ii) Finding Side DA: [1.5 Marks]

$12 + 14 = DA + 15 \implies 26 = DA + 15 \implies$ **DA = 11** cm.

(iii) Tangent Segments on AB (given $\angle B = 90°, r = 4$ cm): [1.5 Marks]

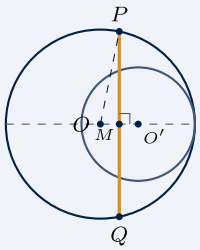
Since $OP \perp AB$, $OQ \perp BC$, $\angle B = 90°$, and $OP = OQ = r = 4$ cm, $OPBQ$ is a square $\implies BP = r = 4$ cm. Therefore $AP = AB - BP = 12 - 4 = 8$ cm.

$DA = \mathbf{11}$ cm, Segments on AB : $BP = \mathbf{4}$ cm, $AP = \mathbf{8}$ cm

Q12. Two circles with centres O and O' of radii 5 cm and 3 cm touch each other internally. If the perpendicular bisector of the segment joining their centres meets the larger circle at points P and Q, find the length of chord PQ. [5]

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let the larger circle have centre O ($R = 5$ cm) and smaller circle have centre O' ($r = 3$ cm). Let PQ intersect OO' at M.



Step 1: Distance Between Centres OO' [1.5 Marks]

Since the circles touch internally, the distance between centres is $OO' = R - r = 5 - 3 = 2$ cm.

Step 2: Midpoint Distance OM [1 Mark]

Since PQ is the perpendicular bisector of OO', $OM = \frac{1}{2}OO' = \frac{2}{2} = 1$ cm and $\angle OMP = 90°$.

Step 3: Applying Pythagoras Theorem in $\triangle OMP$ [2.5 Marks]

Join $OP = R = 5$ cm. In right $\triangle OMP$: $PM = \sqrt{OP^2 - OM^2} = \sqrt{5^2 - 1^2} = \sqrt{24} = 2\sqrt{6}$ cm.

Since perpendicular OM from centre O bisects chord PQ, $PQ = 2PM = 2(2\sqrt{6}) = 4\sqrt{6}$ cm.

Length of chord $PQ = \mathbf{4\sqrt{6}}$ cm ($\approx$ **9.80** cm)

Q13. Security Perimeter Wall Circumscribing a Vault

A security compound builds a four-sided boundary wall $ABCD$ circumscribing a circular bank vault. The wall lengths are $AB = 24$ m, $BC = 27$ m, $CD = 31$ m. Security sensors are positioned along the perimeter.

(i) State the theorem governing the side lengths of a circumscribed quadrilateral. [1]

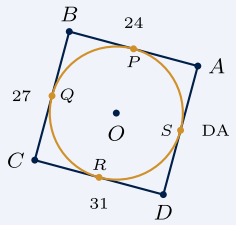
(ii) Calculate the required length of the fourth boundary wall DA. [1]

(iii) Find the total perimeter of the security compound, OR if the vault has radius $r = 10$ m, find the total area of the four triangles formed by connecting the vault centre to the vertices. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Governing Theorem: [1 Mark]

In a quadrilateral circumscribing a circle, sums of opposite sides are equal: $\boxed{\mathbf{AB + CD = BC + DA}}$.



(ii) Length of Fourth Wall DA: [1 Mark]

$24 + 31 = 27 + DA \implies 55 = 27 + DA \implies \boxed{DA = \mathbf{28}\text{ m}}$.

(iii) Main Choice — Total Perimeter of Compound: [2 Marks]

Perimeter $= AB + BC + CD + DA = 24 + 27 + 31 + 28 = \boxed{\mathbf{110}\text{ m}}$.

— OR — (Alternative Choice — Total Area of Triangles, $r = 10$ m): [2 Marks]

Total Area $= \frac{1}{2}r(AB + BC + CD + DA) = \frac{1}{2} \times 10 \times 110 = \boxed{\mathbf{550}\text{ m}^{\mathbf{2}}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★